

3.1: Complex Numbers

Definition. (Imaginary and Complex Numbers)

We can extend the Real numbers by defining:

$$i = \sqrt{-1}.$$

A **complex number** is a number of the form $a + bi$ where

- a is the real part of the complex number, and
- bi is the imaginary part of the complex number.

If $b = 0$, then $a + bi$ is a real number.

If $a = 0$ and $b \neq 0$, then the complex number is called an **imaginary number**.

An imaginary number is an *even* root of a negative number.

$$\begin{array}{ccc} & \nearrow & \nwarrow \\ & 5 + 2i & \\ \text{Real part} & & \text{Imaginary part} \end{array}$$

Example. Express the following complex numbers in standard form:

$$\sqrt{-9} = \sqrt{-1} \sqrt{9} = 3i$$

$$\boxed{0 + 3i}$$

$$5 + \sqrt{-3}$$

$$\boxed{5 + \sqrt{3}i}$$

$$4 - \sqrt{-16}$$

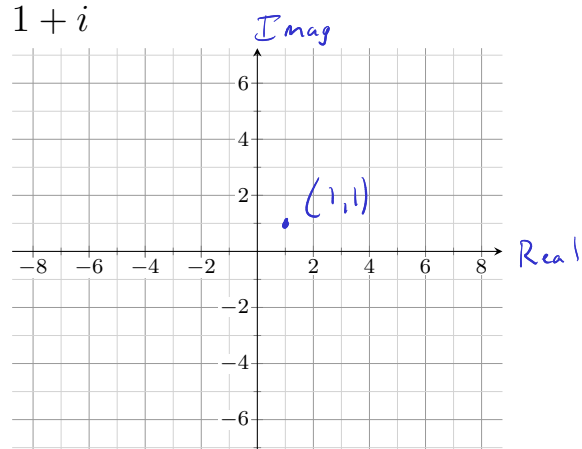
$$\boxed{4 - 4i}$$

$$3 - \sqrt{25}$$

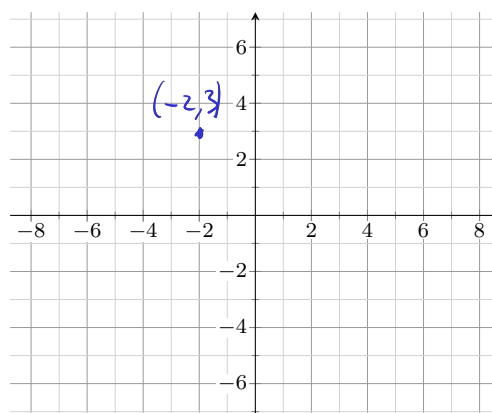
$$3 - 5 = \boxed{-2 + 0i}$$

Example. Graph the following complex numbers:

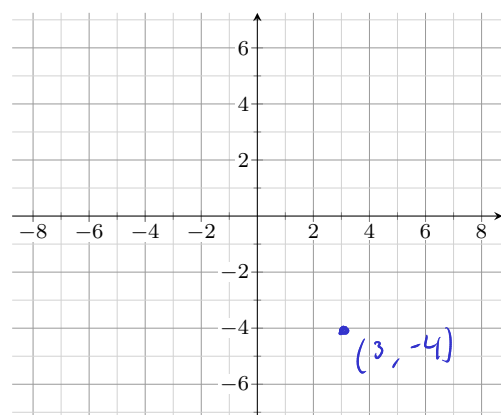
$$1 + i$$



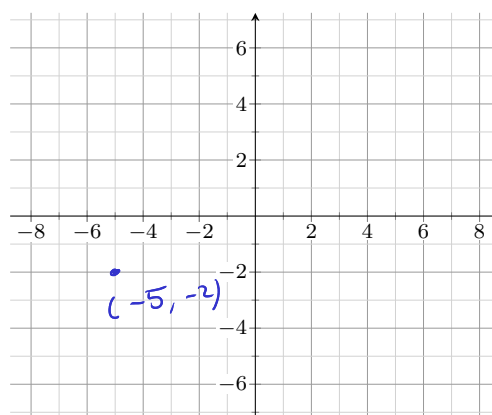
$$-2 + 3i$$



$$3 - 4i$$



$$-5 - 2i$$



Definition. (Addition and Subtraction of complex numbers)

Given complex numbers $a + bi$ and $c + di$,

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

$$(a + bi) - (c + di) = (a + c) - (b + d)i$$

Example. Evaluate the following:

$$\begin{aligned}(3 + 2i) + (1 - 4i) \\&= (3 + 1) + (2 - 4)i \\&= \boxed{4 - 2i}\end{aligned}$$

$$\begin{aligned}(-4 - 2i) - (-1 + 5i) \\&= (-4 - (-1)) + (-2 - 5)i \\&= \boxed{-3 - 7i}\end{aligned}$$

$$\begin{aligned}(2) - (-1 + 2i) \\&= (2 - (-1)) + (-2)i \\&= \boxed{3 - 2i}\end{aligned}$$

$$\begin{aligned}(5i) + (3 + i) \\&= (3) + (5 + 1)i \\&= \boxed{3 + 6i}\end{aligned}$$

Definition. (Multiplying complex numbers)

Given complex numbers $a + bi$ and $c + di$,

$$\begin{aligned}(a + bi) \cdot (c + di) &= ac + adi + bci + bdi^2 \\ &= ac + adi + bci - bd \\ &= (ac - bd) + (ad + bc)i\end{aligned}$$

Example. Evaluate the following:

$$\begin{aligned}(3 + 2i) \cdot (1 - 4i) \\ &= 3 + 2i - 12i - 8\underbrace{i^2}_{-1} \\ &= 3 - 10i - 8(-1) \\ &= \boxed{11 - 10i}\end{aligned}$$

$$\begin{aligned}(-4 - 2i) \cdot (-1 + 5i) \\ &= 4 + 2i - 20i - 10i^2 \\ &= \boxed{14 - 18i}\end{aligned}$$

$$\begin{aligned}(2) \cdot (-1 + 2i) \\ &= \boxed{-2 + 4i}\end{aligned}$$

$$\begin{aligned}(5i) \cdot (3 + i) \\ &= 15i + 5i^2 \\ &= \boxed{-5 + 15i}\end{aligned}$$

Definition. (Dividing complex numbers)

Given complex numbers $a + bi$ and $c + di$, division is done via the *complex conjugate*:

$$\begin{aligned}\frac{c + di}{a + bi} \cdot \frac{a - bi}{a - bi} &= \frac{(c + di)(a - bi)}{(a + bi)(a - bi)} \\ &= \frac{ac - bci + adi - bdi^2}{a^2 - abi + abi - b^2i^2} \\ &= \frac{(ac + bd) + (ad - bc)i}{a^2 + b^2}\end{aligned}$$

Example. Evaluate the following:

$$\frac{3 + 2i}{1 - 4i} \left(\frac{1 + 4i}{1 + 4i} \right) = \frac{3 + 12i + 2i + 8i^2}{1^2 - (4i)^2}$$

$$= \frac{-5 + 14i}{17}$$

$$= \boxed{-\frac{5}{17} + \frac{14}{17}i}$$

$$\frac{-4 - 2i}{-1 + 5i} \left(\frac{-1 - 5i}{-1 - 5i} \right) = \frac{4 + 20i + 2i + 10i^2}{(-1)^2 - (5i)^2}$$

$$= \frac{-6 + 22i}{26}$$

$$= \boxed{-\frac{3}{13} + \frac{11}{13}i}$$

$$\frac{2}{-1 + 2i} \left(\frac{-1 - 2i}{-1 - 2i} \right) = \frac{-2 - 4i}{(-1)^2 - (2i)^2}$$

$$= \frac{-2 - 4i}{5}$$

$$= \boxed{-\frac{2}{5} - \frac{4}{5}i}$$

$$\frac{3 + i}{5i} \left(\frac{-5i}{-5i} \right) = \frac{-15i - 5i^2}{-(5i)^2}$$

$$= \frac{5 - 15i}{25}$$

$$= \boxed{\frac{1}{5} - \frac{3}{5}i}$$

Example. Given $f(x) = x^2 - 5x + 2$, evaluate $f(3 + i)$.

$$\begin{aligned} f(3+i) &= (3+i)^2 - 5(3+i) + 2 \\ &= (9 + 6i + \underbrace{i^2}_{-(-1)}) - 15 - 5i + 2 \\ &= 10 + 6i - 13 - 5i \\ &= \boxed{-3 + i} \end{aligned}$$

Example. Given $f(x) = \frac{2+x}{x+3}$, evaluate $f(10i)$.

$$\begin{aligned} f(10i) &= \frac{2 + 10i}{10i + 3} \left(\frac{3 - 10i}{3 - 10i} \right) = \frac{6 - 20i + 30i - 100i^2}{(3)^2 - (10i)^2} \\ &= \frac{106 + 10i}{109} \\ &= \boxed{\frac{106}{109} + \frac{10}{109}i} \end{aligned}$$

Example. Find i raised to different powers: $i, i^2, i^3, \dots$, then evaluate i^{35} .

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = i^2 \cdot i = (-1)i = -i$$

$$i^4 = i^3 \cdot i = (-i)i = -i^2 = -(-1) = 1$$

$$\Rightarrow i^{35} = i^{32} \cdot i^3 = \underbrace{(i^4)^8}_1 \cdot (-i) = \boxed{-i}$$

multiple
of 4

n	i^n
1	i
2	-1
3	$-i$
4	1
5	i
6	-1
7	$-i$
8	1
$\vdots$	$\vdots$
$\vdots$	$\vdots$

Repeats